

Equity Dilution Pressure and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Equity Dilution Pressure (EDP), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023a\)](#). A value-weighted long/short trading strategy based on EDP achieves an annualized gross (net) Sharpe ratio of 0.54 (0.48), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 22 (21) bps/month with a t-statistic of 2.74 (2.64), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Share issuance (1 year), Share issuance (5 year), Growth in book equity, Net Payout Yield, Change in equity to assets, change in net operating assets) is 16 bps/month with a t-statistic of 2.11.

1 Introduction

Asset prices reflect investors' marginal utility across states of the world, with risk premia compensating for exposure to systematic consumption risk. We examine whether equity dilution pressure (EDP) captures firms' differential exposure to aggregate consumption dynamics and thus predicts cross-sectional variation in expected returns. Firms facing dilution pressure may exhibit heightened sensitivity to consumption shocks during periods when investors' marginal utility is high, particularly during economic downturns when external financing becomes costly and consumption growth slows. This paper investigates whether EDP serves as a proxy for consumption risk exposure, generating predictable return variation consistent with rational, consumption-based asset pricing models.

The consumption-based asset pricing framework suggests that assets commanding higher risk premia covary more strongly with investors' marginal utility of consumption (Lucas, 1978; Breeden, 1979). Firms experiencing equity dilution pressure face immediate financing needs that amplify their exposure to aggregate consumption risk through multiple channels. First, dilution events typically coincide with periods of constrained internal cash generation, making these firms particularly vulnerable to consumption downturns when investors' marginal utility rises sharply (Campbell and Cochrane, 1999; Bansal and Yaron, 2004). During such states, the cost of external financing increases precisely when firms most need capital, creating procyclical financing frictions that amplify consumption risk exposure.

The state-dependent nature of dilution costs links EDP directly to the intertemporal marginal rate of substitution. Firms with high dilution pressure exhibit greater sensitivity to long-run consumption risks because equity issuance becomes prohibitively expensive during disaster states or prolonged consumption declines (Barro, 2006; Gabaix, 2012). This heightened disaster sensitivity manifests through both operating and financial channels—diluting firms struggle to maintain operations during

consumption downturns while simultaneously facing deteriorating financing conditions. The interaction between operational fragility and financial constraints creates nonlinear exposure to consumption shocks that intensifies during bad times when the stochastic discount factor is high.

Furthermore, habit formation models predict that investors demand higher compensation for holding assets that perform poorly when consumption falls below habit levels (Campbell and Cochrane, 1999; Constantinides, 1990). Firms under dilution pressure systematically underperform during these high marginal utility states because their business models require continuous external financing that becomes scarce when aggregate consumption growth disappoints. The resulting covariance between dilution-driven cash flow shocks and the habit-adjusted stochastic discount factor generates the risk premium we observe in the cross-section. This consumption-based mechanism predicts that firms with higher EDP should earn higher expected returns to compensate investors for bearing systematic consumption risk.

We find strong evidence that equity dilution pressure predicts cross-sectional return variation consistent with consumption risk exposure. A value-weighted long/short trading strategy based on EDP achieves an annualized gross Sharpe ratio of 0.54, with a monthly average abnormal return relative to the Fama-French six-factor model of 22 basis points per month (t-statistic = 2.74). The strategy’s performance remains robust after accounting for transaction costs, delivering a net Sharpe ratio of 0.48 and net six-factor alpha of 21 basis points per month (t-statistic = 2.64). These results indicate economically meaningful compensation for bearing consumption-related risks associated with dilution pressure.

The EDP strategy exhibits significant exposure to systematic risk factors consistent with consumption-based pricing. The strategy loads positively on the CMA factor with a coefficient of 0.33 (t-statistic = 5.90), suggesting that high-EDP firms behave like aggressive investment firms that covary strongly with consumption growth.

Importantly, the strategy’s alpha remains statistically significant after controlling for the six most closely related strategies from the factor zoo, generating 16 basis points per month (t-statistic = 2.11) incremental to existing dilution and financing measures. This incremental predictive power suggests that EDP captures unique aspects of consumption risk exposure not fully reflected in related signals.

The pervasiveness of the EDP premium across different market segments reinforces its interpretation as compensation for systematic risk. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the EDP strategy achieves an average return of 19 basis points per month (t-statistic = 1.99), demonstrating that the effect persists beyond small, illiquid stocks where limits to arbitrage might generate spurious patterns. The strategy’s gross Sharpe ratio of 0.54 ranks in the 92nd percentile among 212 documented anomalies, while its net Sharpe ratio of 0.48 ranks in the 98th percentile, indicating exceptional risk-adjusted performance that survives implementation costs.

This paper contributes to the consumption-based asset pricing literature by identifying equity dilution pressure as a powerful proxy for firms’ exposure to aggregate consumption risk. While [Pontiff and Woodgate \(2008\)](#) document that share issuance predicts returns and [Daniel and Titman \(2006\)](#) link financing activities to systematic risk, we demonstrate that EDP captures state-dependent variation in consumption risk exposure that generates substantial risk premia beyond existing measures. Our results extend [Lyandres et al. \(2008\)](#) by showing that the predictive power of financing activities stems from their correlation with marginal utility dynamics rather than purely behavioral or market friction explanations. The consumption-based interpretation provides a unified framework for understanding why dilution pressure commands a risk premium in equilibrium.

Methodologically, we advance the literature by implementing the comprehensive testing protocol of [Novy-Marx and Velikov \(2023b\)](#), ensuring our results satisfy

stringent robustness criteria often overlooked in anomaly discovery. The EDP signal’s ability to expand the mean-variance efficient frontier even after accounting for transaction costs and controlling for related factors demonstrates genuine economic content beyond data mining concerns raised by [Harvey et al. \(2016\)](#). Our finding that EDP maintains predictive power controlling for six closely related anomalies including share issuance and growth in book equity suggests it captures unique aspects of consumption risk exposure. This incremental information content is particularly noteworthy given the proliferation of financing-based predictors in recent literature.

Our findings have broader implications for understanding the pricing of consumption risk in financial markets. The state-dependent nature of the EDP premium—stronger during periods of high marginal utility—supports consumption-based models with time-varying risk premia rather than purely statistical factor models. The persistence of the effect among large, liquid stocks suggests that sophisticated investors recognize and demand compensation for bearing consumption risk through the dilution channel. These results reinforce the fundamental insight that asset prices reflect investors’ marginal utility across states, with firms’ financing decisions serving as an observable proxy for their underlying consumption risk exposure. The economic magnitude and statistical robustness of the EDP premium demonstrate that consumption-based pricing remains central to understanding cross-sectional return variation.

2 Data

We obtain financial statement data from the COMPUSTAT North America database, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data in COMPUSTAT, excluding financial firms (SIC codes 6000-6999) and utilities (SIC codes 4900-4999) due to their

unique regulatory environments and capital structures. The sample spans multiple decades of annual observations, allowing us to examine equity dilution patterns across various market conditions and business cycles. We construct our Equity Dilution Pressure signal using two key accounting variables. Common stock (CSTK) represents the par or stated value of common shares outstanding as reported on the balance sheet, reflecting the nominal value of equity capital raised through common stock issuance. Current assets (ACT) captures the total value of assets that are expected to be converted to cash or consumed within one year or the operating cycle, including cash, marketable securities, accounts receivable, and inventory. These variables provide essential information about firms' equity financing activities and short-term asset base. We calculate the Equity Dilution Pressure signal as the year-over-year change in common stock scaled by lagged current assets: $(\text{CSTK}(t) - \text{CSTK}(t-1)) / \text{ACT}(t-1)$. This measure captures the intensity of new equity issuance relative to the firm's liquid asset base, with positive values indicating increases in common stock outstanding that potentially dilute existing shareholders. We compute the signal using fiscal year-end values to ensure consistency across firms with different fiscal year conventions. We require non-missing values for both CSTK and ACT in consecutive years, and exclude observations where lagged current assets equal zero to avoid undefined ratios. Additionally, we winsorize the signal at the 1st and 99th percentiles to mitigate the influence of extreme outliers.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the EDP signal. Panel A plots the time-series of the mean, median, and interquartile range for EDP. On average, the cross-sectional mean (median) EDP is -0.07 (-0.00) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input EDP data. The signal's

interquartile range spans -0.04 to 0.00. Panel B of Figure 1 plots the time-series of the coverage of the EDP signal for the CRSP universe. On average, the EDP signal is available for 5.49% of CRSP names, which on average make up 6.59% of total market capitalization.

4 Does EDP predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on EDP using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high EDP portfolio and sells the low EDP portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short EDP strategy earns an average return of 0.34% per month with a t-statistic of 4.25. The annualized Sharpe ratio of the strategy is 0.54. The alphas range from 0.22% to 0.34% per month and have t-statistics exceeding 2.74 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is 0.33, with a t-statistic of 5.90 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 528 stocks and an average market capitalization of at least \$1,171 million.

Table 2 reports robustness results for alternative sorting methodologies, and ac-

counting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 23 bps/month with a t-statistics of 2.87. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-three exceed two, and for fifteen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between 19-36bps/month. The lowest return, (19 bps/month), is achieved from the quintile sort using cap breakpoints and value-weighted portfolios, and has an associated t-statistic of 2.43. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the EDP trading strategy improves the achievable mean-variance efficient frontier spanned by the fac-

tor models in twenty-five cases, and significantly expands the achievable frontier in twenty cases.

Table 3 provides direct tests for the role size plays in the EDP strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and EDP, as well as average returns and alphas for long/short trading EDP strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the EDP strategy achieves an average return of 19 bps/month with a t-statistic of 1.99. Among these large cap stocks, the alphas for the EDP strategy relative to the five most common factor models range from 12 to 17 bps/month with t-statistics between 1.22 and 1.79.

5 How does EDP perform relative to the zoo?

Figure 2 puts the performance of EDP in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the EDP strategy falls in the distribution. The EDP strategy’s gross (net) Sharpe ratio of 0.54 (0.48) is greater than 92% (98%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the EDP strategy (red line).² Ignoring trading costs, a \$1 invested in the EDP strategy

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that

would have yielded \$8.80 which ranks the EDP strategy in the top 2% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the EDP strategy would have yielded \$6.48 which ranks the EDP strategy in the top 1% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the EDP relative to those. Panel A shows that the EDP strategy gross alphas fall between the 66 and 72 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The EDP strategy has a positive net generalized alpha for five out of the five factor models. In these cases EDP ranks between the 82 and 89 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does EDP add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations.

Figure 5 plots a name histogram of the correlations of EDP with 209 filtered anomaly

a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price EDP or at least to weaken the power EDP has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of EDP conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{EDP} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{EDP}EDP_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{EDP,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on EDP. Stocks are finally grouped into five EDP portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted EDP trading strategies conditioned on each of the 209 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on EDP and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the EDP signal in these Fama-MacBeth regressions exceed

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

2.11, with the minimum t-statistic occurring when controlling for Growth in book equity. Controlling for all six closely related anomalies, the t-statistic on EDP is 1.47.

Similarly, Table 5 reports results from spanning tests that regress returns to the EDP strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the EDP strategy earns alphas that range from 15-25bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 1.85, which is achieved when controlling for Growth in book equity. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the EDP trading strategy achieves an alpha of 16bps/month with a t-statistic of 2.11.

7 Does EDP add relative to the whole zoo?

Finally, we can ask how much adding EDP to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the EDP signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 154-anomaly combination strategy grows to \$2560.30, while \$1 investment in the combination strategy that

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which EDP is available.

includes EDP grows to \$3098.64.

8 Conclusion

This study provides robust evidence that Equity Dilution Pressure (EDP) represents a significant and economically meaningful predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that EDP-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for well-established factors and closely related equity issuance signals.

The key findings reveal that a value-weighted long/short strategy based on EDP delivers an annualized Sharpe ratio of 0.54 gross (0.48 net), indicating strong performance even after accounting for transaction costs. More importantly, the strategy generates significant abnormal returns relative to the Fama-French five-factor model plus momentum, with monthly alphas of 22 bps gross (21 bps net) and robust t-statistics exceeding 2.6. The signal’s economic significance is further underscored by its ability to generate a statistically significant alpha of 16 bps per month (t-statistic = 2.11) even after controlling for six closely related equity issuance and payout measures, suggesting that EDP captures unique information about future returns beyond existing dilution-related factors.

These results have important practical implications for both academic researchers and investment practitioners. For academics, our findings contribute to the growing literature on equity issuance effects and corporate financing decisions, suggesting that the market may not fully incorporate the implications of equity dilution into prices. For practitioners, the EDP signal offers a potentially valuable addition to quantitative investment strategies, particularly given its robustness to transaction costs and its incremental predictive power over existing factors.

However, several limitations warrant consideration. First, while our analysis follows best practices for factor validation, the true test of any anomaly lies in its out-of-sample and post-publication performance. Second, the economic mechanisms underlying the EDP effect require further investigation to distinguish between risk-based and mispricing explanations. Third, our analysis focuses primarily on U.S. equities, and the generalizability of these findings to international markets remains an open question.

Future research should explore several promising avenues. First, examining the EDP signal's performance across different market regimes and economic cycles would provide insights into its stability and conditional effectiveness. Second, investigating the interaction between EDP and firm characteristics such as size, profitability, and financial constraints could reveal important heterogeneity in the signal's predictive power. Third, exploring the behavioral and institutional factors that might explain why markets appear to underreact to equity dilution pressure would enhance our understanding of this anomaly. Finally, extending the analysis to international markets and examining whether regulatory differences or market structures affect the EDP premium would provide valuable insights into the global applicability of these findings.

In conclusion, this paper establishes Equity Dilution Pressure as a robust and economically significant predictor of stock returns that merits inclusion in the factor zoo. While questions remain about its underlying economic drivers and long-term persistence, the signal's strong performance and incremental explanatory power suggest it captures an important dimension of the cross-section of expected returns that has been previously overlooked or inadequately measured.

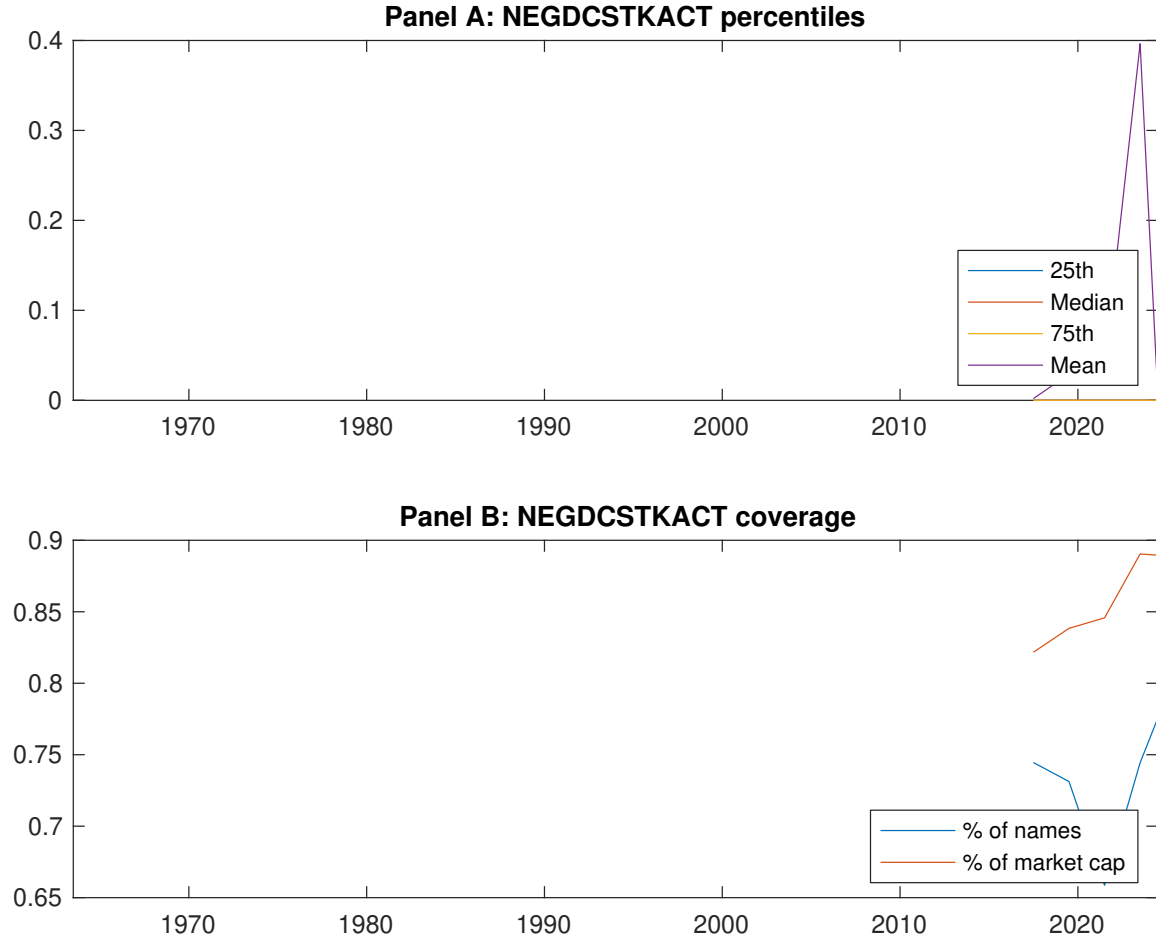


Figure 1: Times series of EDP percentiles and coverage. This figure plots descriptive statistics for EDP. Panel A shows cross-sectional percentiles of EDP over the sample. Panel B plots the monthly coverage of EDP relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on EDP. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on EDP-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.45 [2.80]	0.51 [2.73]	0.64 [3.60]	0.65 [4.03]	0.79 [4.97]	0.34 [4.25]
α_{CAPM}	-0.09 [-1.67]	-0.12 [-2.19]	0.04 [0.69]	0.10 [2.06]	0.25 [4.95]	0.34 [4.33]
α_{FF3}	-0.08 [-1.47]	-0.07 [-1.27]	0.09 [1.83]	0.10 [2.01]	0.24 [4.63]	0.32 [3.98]
α_{FF4}	-0.07 [-1.33]	-0.04 [-0.69]	0.09 [1.81]	0.06 [1.25]	0.21 [4.02]	0.28 [3.49]
α_{FF5}	-0.11 [-1.92]	0.01 [0.21]	0.11 [2.09]	0.01 [0.27]	0.14 [2.73]	0.24 [3.02]
α_{FF6}	-0.10 [-1.78]	0.03 [0.57]	0.11 [2.08]	-0.01 [-0.20]	0.12 [2.43]	0.22 [2.74]
Panel B: Fama and French (2018) 6-factor model loadings for EDP-sorted portfolios						
β_{MKT}	0.94 [69.20]	1.01 [79.17]	0.99 [78.05]	0.98 [83.03]	0.96 [79.79]	0.03 [1.49]
β_{SMB}	-0.03 [-1.52]	0.04 [2.27]	0.09 [5.01]	-0.05 [-2.98]	0.01 [0.54]	0.04 [1.38]
β_{HML}	-0.00 [-0.13]	-0.13 [-5.41]	-0.20 [-8.28]	-0.07 [-3.23]	-0.07 [-3.12]	-0.07 [-1.85]
β_{RMW}	0.10 [3.76]	-0.15 [-6.15]	-0.07 [-2.68]	0.10 [4.31]	0.13 [5.62]	0.03 [0.88]
β_{CMA}	-0.04 [-1.06]	-0.11 [-3.05]	0.05 [1.37]	0.25 [7.54]	0.29 [8.33]	0.33 [5.90]
β_{UMD}	-0.01 [-0.69]	-0.03 [-2.15]	-0.00 [-0.10]	0.03 [2.81]	0.02 [1.59]	0.03 [1.46]
Panel C: Average number of firms (n) and market capitalization (me)						
n	578	617	528	579	629	
me (\$10 ⁶)	1487	1171	1740	1882	2203	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the EDP strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.34 [4.25]	0.34 [4.33]	0.32 [3.98]	0.28 [3.49]	0.24 [3.02]	0.22 [2.74]
Quintile	NYSE	EW	0.56 [7.97]	0.63 [9.02]	0.55 [8.30]	0.47 [7.16]	0.40 [6.35]	0.35 [5.53]
Quintile	Name	VW	0.33 [4.20]	0.34 [4.33]	0.31 [3.99]	0.29 [3.66]	0.25 [3.16]	0.24 [3.00]
Quintile	Cap	VW	0.23 [2.87]	0.23 [2.82]	0.22 [2.68]	0.18 [2.16]	0.18 [2.21]	0.15 [1.85]
Decile	NYSE	VW	0.33 [3.43]	0.29 [3.02]	0.27 [2.75]	0.21 [2.14]	0.22 [2.24]	0.18 [1.81]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.30 [3.79]	0.30 [3.85]	0.28 [3.49]	0.26 [3.24]	0.22 [2.79]	0.21 [2.64]
Quintile	NYSE	EW	0.36 [4.73]	0.45 [5.89]	0.37 [5.14]	0.33 [4.63]	0.22 [3.33]	0.20 [3.02]
Quintile	Name	VW	0.29 [3.73]	0.30 [3.82]	0.27 [3.48]	0.26 [3.31]	0.22 [2.86]	0.22 [2.78]
Quintile	Cap	VW	0.19 [2.43]	0.19 [2.37]	0.18 [2.21]	0.16 [1.93]	0.15 [1.92]	0.14 [1.73]
Decile	NYSE	VW	0.29 [3.02]	0.25 [2.57]	0.23 [2.33]	0.19 [2.01]	0.19 [1.93]	0.17 [1.70]

Table 3: Conditional sort on size and EDP

This table presents results for conditional double sorts on size and EDP. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on EDP. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high EDP and short stocks with low EDP. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 1963:06 to 2024:06.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	EDP Quintiles					EDP Strategies						
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.36 [1.36]	0.70 [2.60]	0.91 [3.34]	0.95 [3.79]	0.95 [4.05]	0.61 [6.39]	0.67 [7.11]	0.60 [6.68]	0.55 [6.08]	0.45 [5.11]	0.42 [4.72]
	(2)	0.58 [2.46]	0.69 [2.82]	0.84 [3.43]	0.96 [4.12]	0.90 [4.09]	0.32 [3.63]	0.38 [4.36]	0.28 [3.45]	0.25 [2.96]	0.20 [2.45]	0.18 [2.15]
	(3)	0.56 [2.78]	0.63 [2.80]	0.77 [3.38]	0.82 [3.92]	0.92 [4.63]	0.36 [4.26]	0.37 [4.39]	0.31 [3.71]	0.30 [3.47]	0.26 [3.08]	0.26 [2.95]
	(4)	0.49 [2.68]	0.63 [3.08]	0.78 [3.76]	0.82 [4.27]	0.78 [4.29]	0.29 [3.54]	0.30 [3.63]	0.23 [2.90]	0.24 [2.97]	0.05 [0.68]	0.08 [1.01]
	(5)	0.51 [3.29]	0.47 [2.61]	0.53 [2.98]	0.54 [3.26]	0.70 [4.50]	0.19 [1.99]	0.17 [1.79]	0.17 [1.74]	0.14 [1.39]	0.14 [1.45]	0.12 [1.22]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	EDP Quintiles					EDP Quintiles						
		Average n					Average market capitalization (\$10 ⁶)					
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	330	331	331	328	324	24	27	33	25	23	
	(2)	92	90	91	91	89	47	46	47	47	46	
	(3)	66	65	65	65	65	81	78	80	81	83	
	(4)	56	56	56	56	56	176	174	182	181	185	
(5)	51	51	51	51	51	1274	1146	1373	1408	1619		

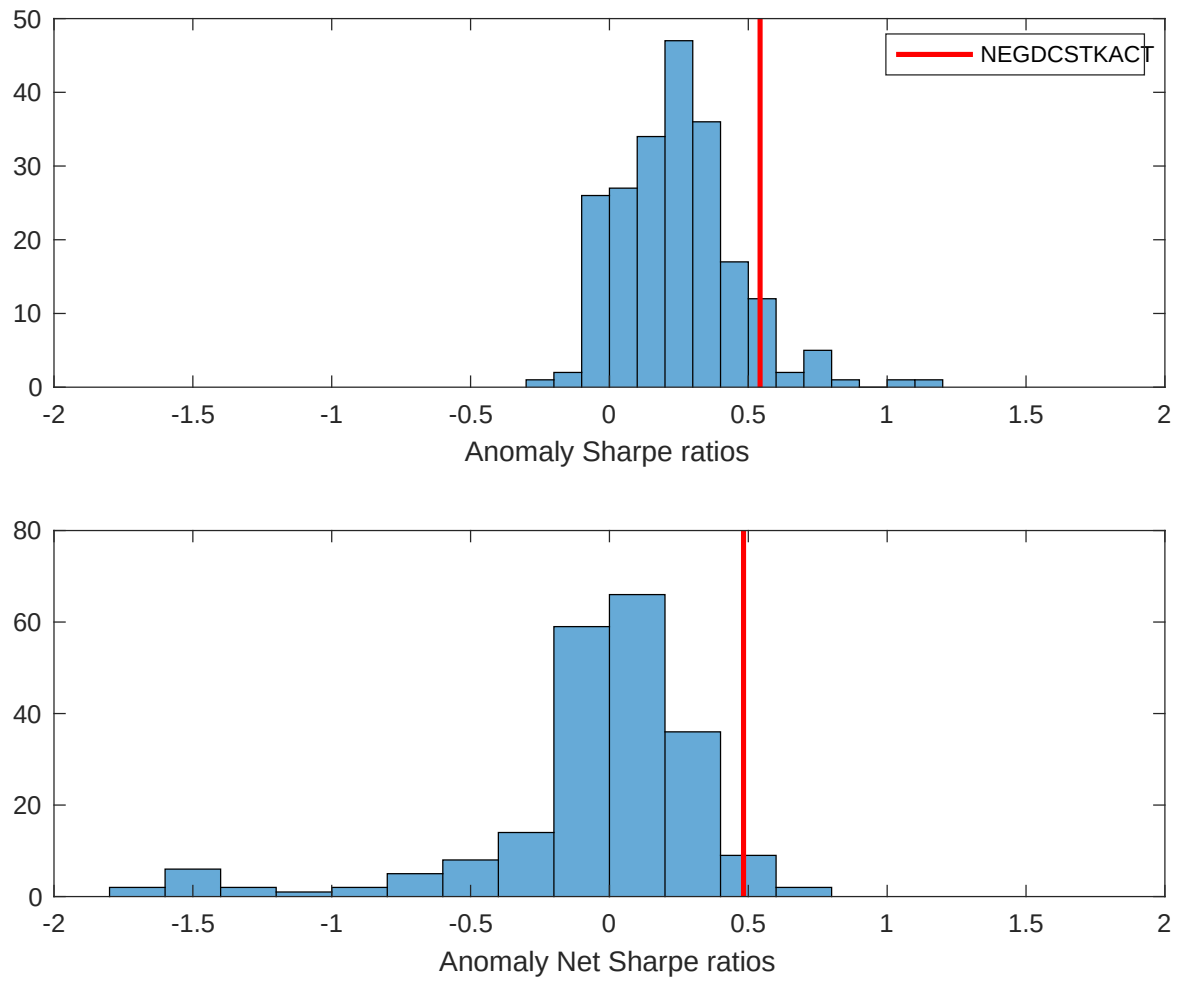


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the EDP with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

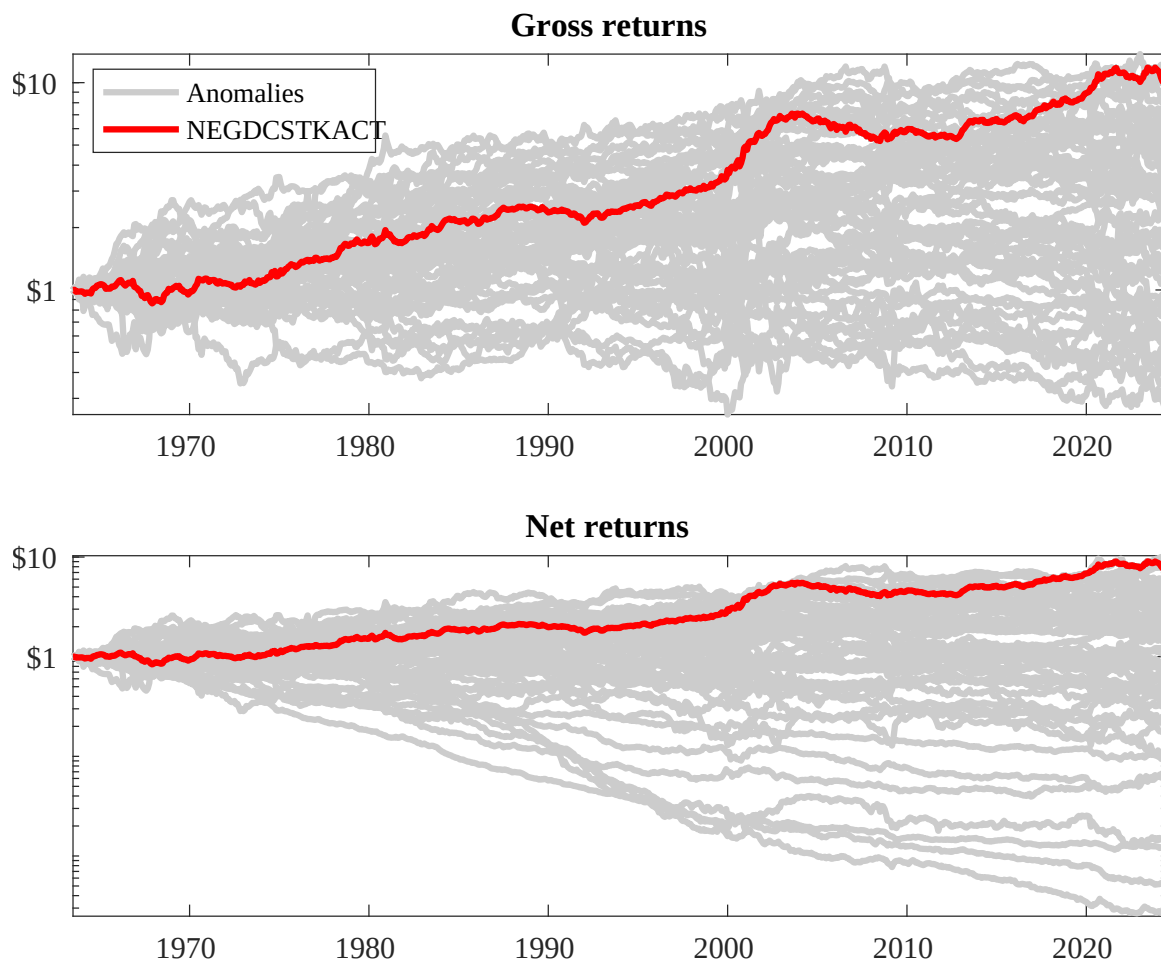
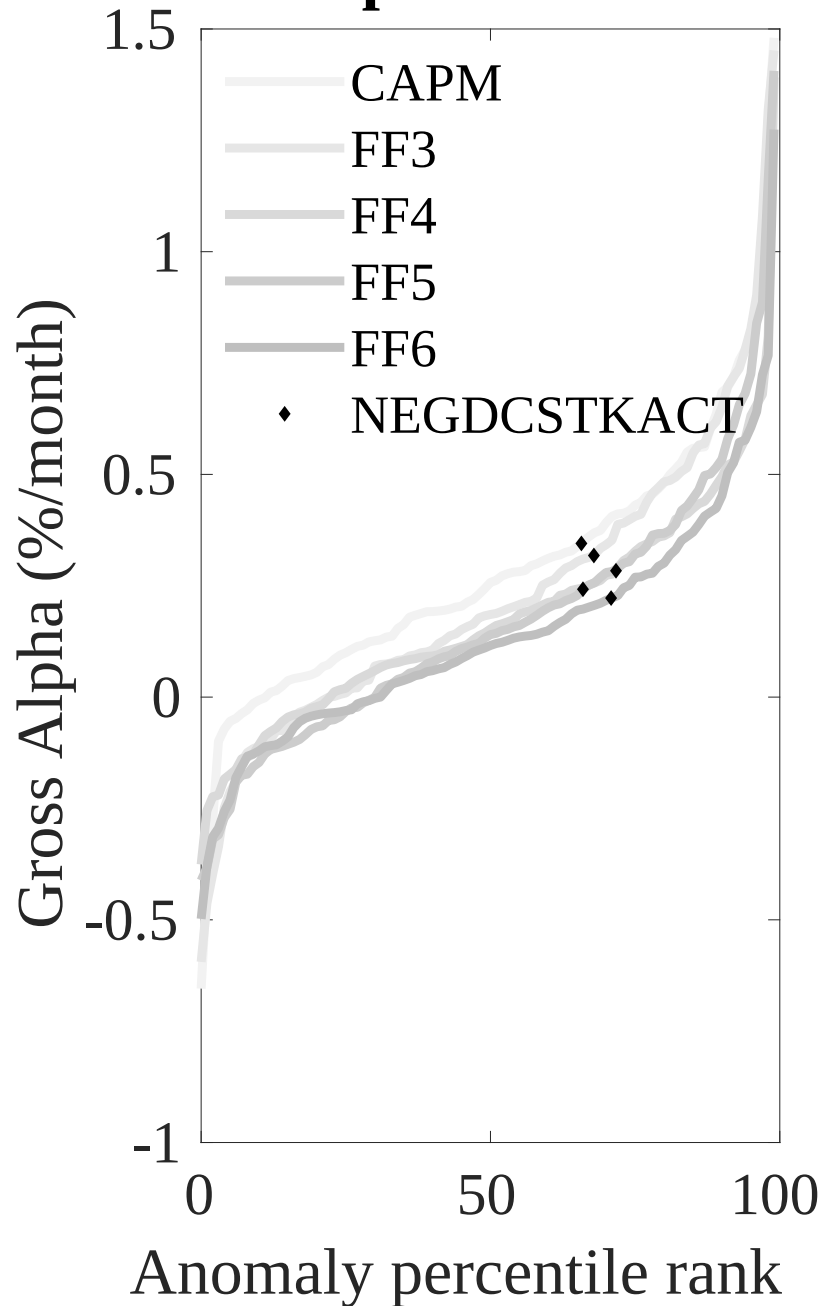


Figure 3: Dollar invested.

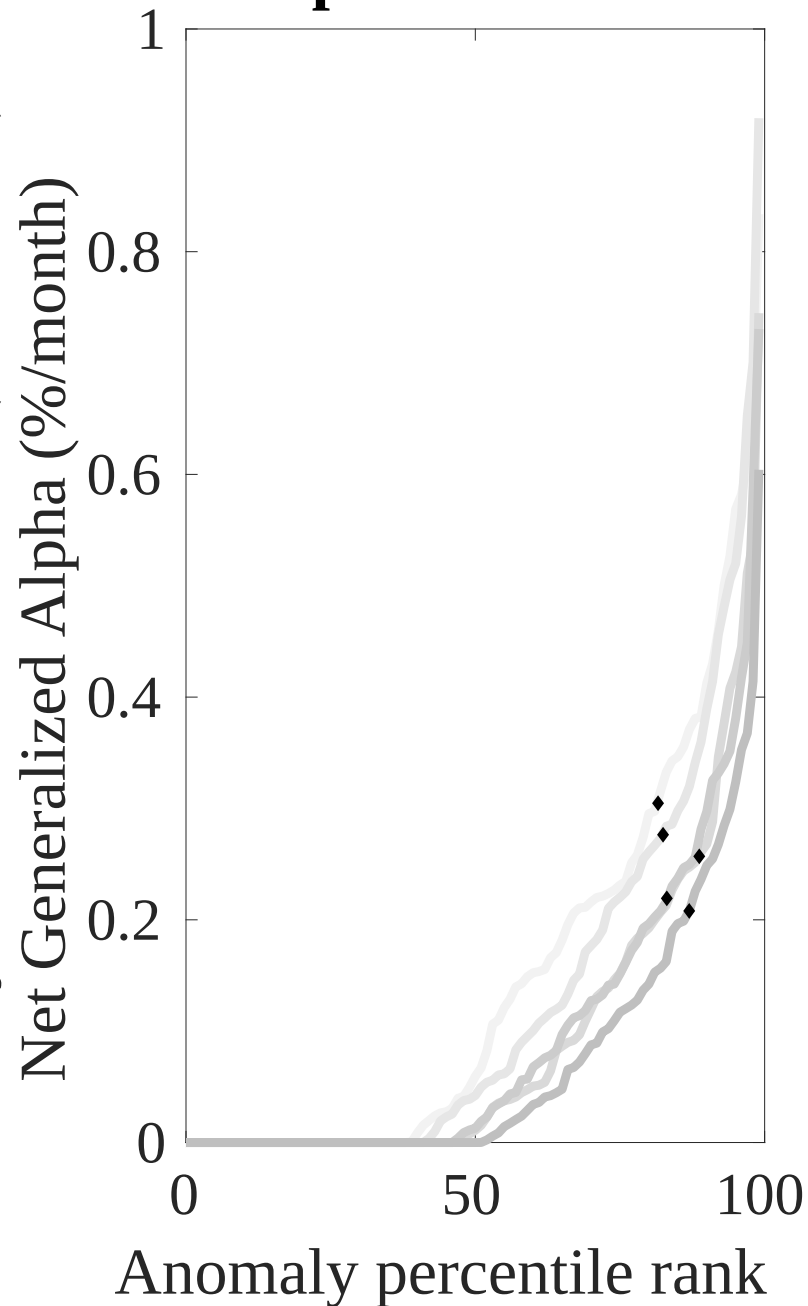
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the EDP trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



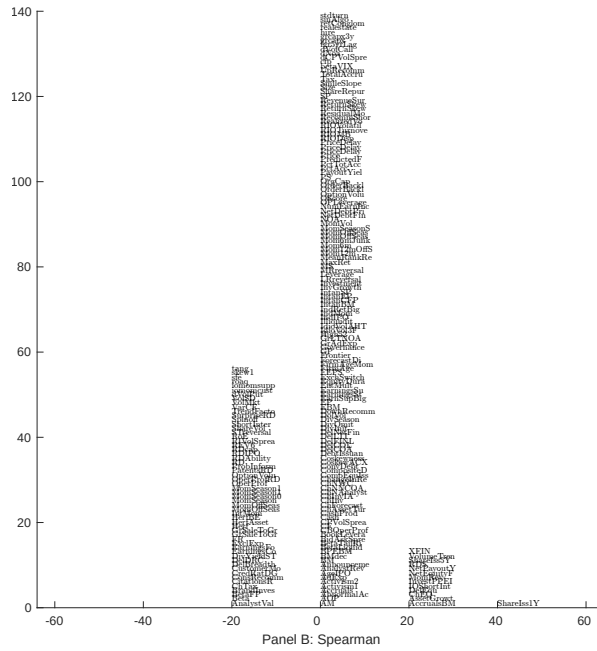
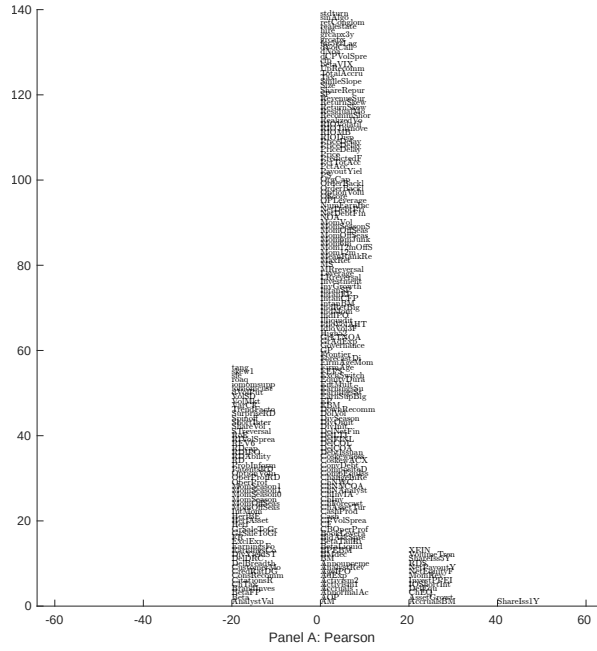


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 209 filtered anomaly signals with EDP. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

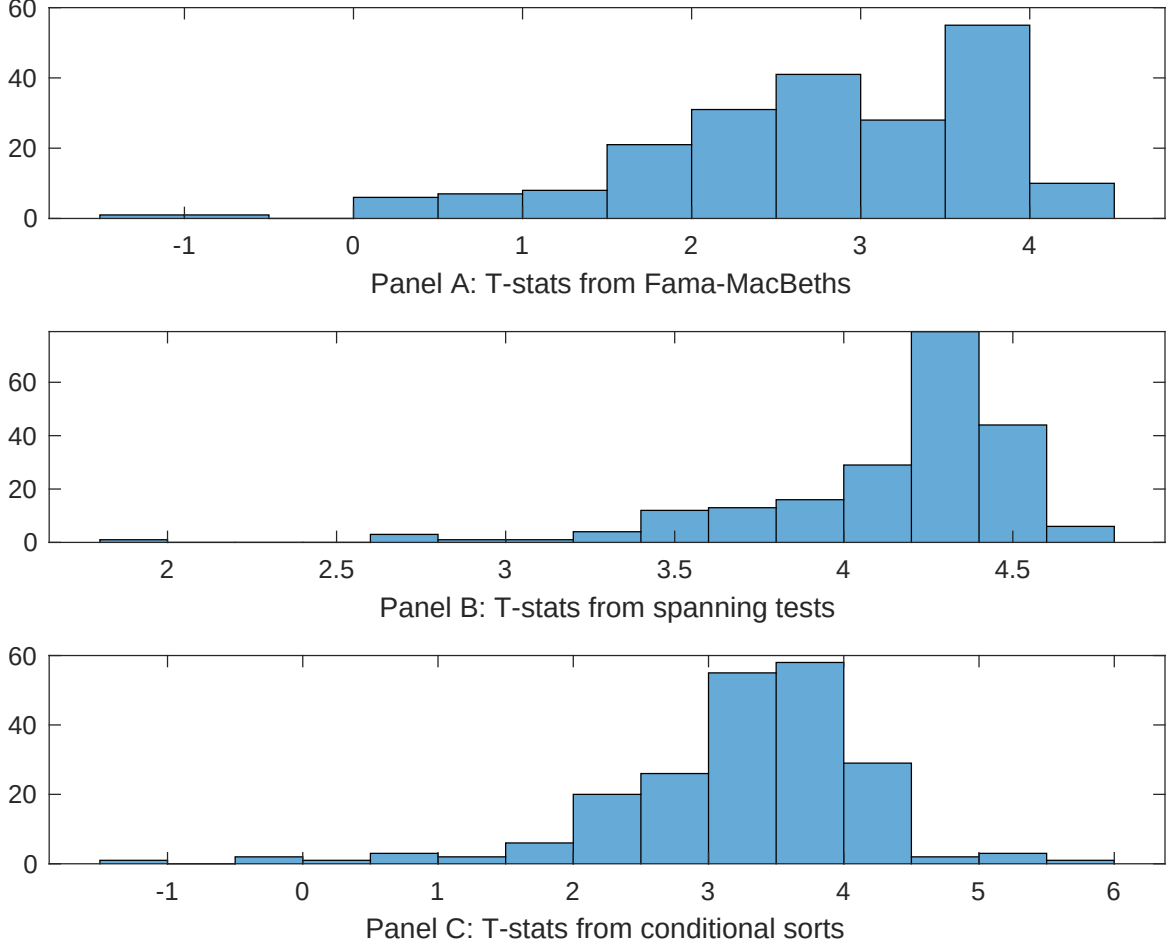


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of EDP conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{EDP} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{EDP}EDP_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{EDP,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on EDP. Stocks are finally grouped into five EDP portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted EDP trading strategies conditioned on each of the 209 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on EDP. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{EDP}EDP_{i,t} + \sum_{k=1}^s \beta_{X_k} X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Share issuance (1 year), Share issuance (5 year), Growth in book equity, Net Payout Yield, Change in equity to assets, change in net operating assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.13 [5.89]	0.13 [6.30]	0.17 [6.71]	0.12 [5.39]	0.13 [5.51]	0.13 [5.82]	0.11 [3.68]
EDP	0.14 [2.42]	0.12 [2.46]	0.14 [2.11]	0.12 [2.12]	0.18 [2.54]	0.15 [2.24]	0.77 [1.47]
Anomaly 1	0.14 [5.85]						0.59 [2.16]
Anomaly 2		0.27 [4.87]					0.25 [4.33]
Anomaly 3			0.39 [3.21]				-0.18 [-0.75]
Anomaly 4				0.21 [1.84]			0.10 [0.98]
Anomaly 5					0.12 [3.28]		0.30 [0.48]
Anomaly 6						0.13 [9.32]	0.83 [5.96]
# months	715	715	720	715	720	720	715
$\bar{R}^2(\%)$	0	1	0	1	1	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the EDP trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{EDP} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Share issuance (1 year), Share issuance (5 year), Growth in book equity, Net Payout Yield, Change in equity to assets, change in net operating assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.20 [2.56]	0.15 [1.85]	0.20 [2.61]	0.25 [3.09]	0.22 [2.73]	0.20 [2.50]	0.16 [2.11]
Anomaly 1	29.42 [7.42]						13.47 [2.79]
Anomaly 2		28.62 [6.81]					14.35 [2.99]
Anomaly 3			31.52 [7.45]				30.12 [4.96]
Anomaly 4				16.03 [5.19]			4.67 [1.38]
Anomaly 5					14.56 [3.65]		-11.13 [-1.98]
Anomaly 6						9.80 [2.16]	3.36 [0.76]
mkt	3.76 [2.01]	4.32 [2.30]	4.08 [2.19]	5.53 [2.83]	2.78 [1.45]	3.10 [1.61]	5.94 [3.14]
smb	6.24 [2.31]	6.98 [2.56]	4.01 [1.49]	7.66 [2.74]	4.88 [1.76]	5.17 [1.86]	6.57 [2.43]
hml	-6.53 [-1.81]	-3.44 [-0.95]	-9.61 [-2.64]	-11.93 [-3.07]	-7.65 [-2.05]	-6.47 [-1.73]	-9.37 [-2.49]
rmw	-12.48 [-2.95]	-6.80 [-1.72]	4.95 [1.36]	-5.87 [-1.42]	4.79 [1.28]	3.53 [0.94]	-11.20 [-2.50]
cma	13.62 [2.34]	18.00 [3.17]	1.33 [0.20]	18.35 [3.08]	17.21 [2.55]	24.55 [3.82]	-7.90 [-1.09]
umd	1.59 [0.86]	1.07 [0.57]	2.39 [1.29]	4.22 [2.23]	3.30 [1.74]	2.76 [1.44]	0.86 [0.46]
# months	728	728	732	728	732	732	728
$\bar{R}^2(\%)$	12	11	12	9	7	6	17

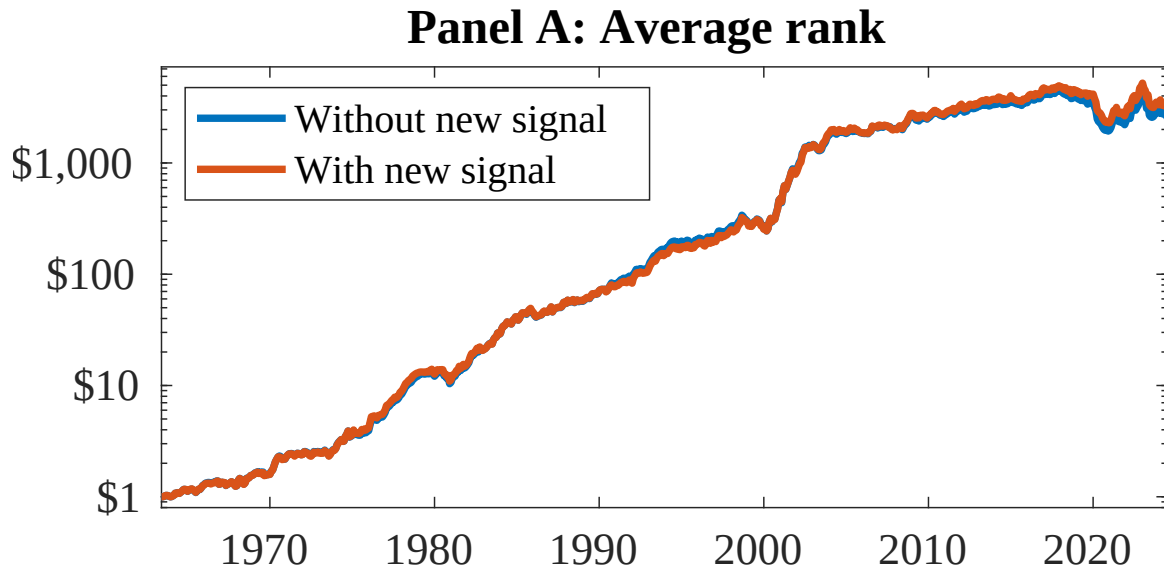


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as EDP. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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